Final Project Submission Template

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Lesson name : Internet and Web Programming (SEN204)

Project Title: GDF.Estate: Real Estate Property Finder

Research Question / Problem Statement: Finding a house online can be confusing if the website has too much static information. My question is: How can I build a simple and interactive web interface that lets users search for properties and filter them by city and status (sale or rent) easily?

Objective of the Project: The goal of this project is to use HTML, CSS, and JavaScript to make a functional real estate webpage. I want to allow users to search and see the results instantly on the screen without needing to refresh the page.

Expected Outcome: A working website with a responsive layout . It will have a search bar , dropdown filters, and a horizontal image slider, all controlled by JavaScript.

Approach to Problem Solving: I divided my project into three parts to keep the code clean:

- **Input:** The user types text in the search bar or selects a city and property status from the dropdown menus.
- **Processing:** My JavaScript code listens to these inputs. I saved my properties in an Array of Objects. The code uses the .filter() method to check which properties match the user's choices.
- **Output:** The JavaScript updates the HTML DOM to show only the property cards that match. I also created a slider to show VIP properties.

Example Control Flow:

1. Start the program (User opens the website).
2. Display the main menu (Search bar, VIP Slider, and all property cards).
3. User selects an option (e.g., types "Villa" or selects "Istanbul").
4. Processing:
    - Read the input values.
    - Loop through the properties array.
    - If the property matches the text, city, AND status: keep it.
5. Output: Clear the old grid and display the new filtered cards.
6. Repeat every time the user changes a filter.

Technical Implementation

Programming Language: HTML, CSS, and JavaScript

```javascript
// Function to filter properties when the user types or selects options
function filterProperties() {
    const searchTerm = searchInput.value.toLowerCase();
    const selectedCity = cityFilter.value;
    const selectedStatus = statusFilter.value;

    // Use the filter method to keep only matching properties
    const filtered = properties.filter(prop => {
        const matchesSearch = prop.name.toLowerCase().includes(searchTerm);
        const matchesCity = selectedCity === 'all' || prop.city === selectedCity;
        const matchesStatus = selectedStatus === 'all' || prop.status === selectedStatus;

        // It must match all 3 conditions to be shown
        return matchesSearch && matchesCity && matchesStatus;
    });

    displayProperties(filtered); // Update the screen with the new list
}
```

Explanation:

In this code, I get the values from the text box and the dropdowns. Then, I use the JavaScript .filter() method on my array of properties. Inside the filter, I check

three things: if the property name contains the search text, if the city matches the dropdown, and if the status matches. If a property passes all three checks, it stays in the filtered array. Finally, I call the displayProperties() function to draw the updated list of cards on the screen.